ARTEFACT SELECTION PROCESS — STAGE 1 TECHNICAL CASE

Empório da Música — AI Customer Support Agent

A Production-Grade, Full-Stack ReAct & Vector RAG Architecture

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1. Executive Summary & Problem Formulation

Empório da Música is a premier musical instruments store established in 2008 in Campo Grande/MS. The human customer service team was overburdened with high-volume repetitive queries regarding product pricing, inventory availability, order status tracking, and store policies (returns, warranties, operating hours).

To solve this challenge, I designed and developed a **full-stack, autonomous AI agent in**

Python featuring an interactive **React 19 web application**, a **Rich terminal CLI**, and a robust hybrid orchestration engine combining **Google Gemini (Tool Calling)**, **Qdrant (Vector RAG)**, and **SQLite / SQLAlchemy (Deterministic Relational Queries)**.

> **Key Mandatory Technical Constraint:** Python as the primary language.
>
> **Architecture Delivered:** 100% Python backend & AI engine (FastAPI + SQLAlchemy + Qdrant + Google GenAI SDK) paired with a modern, responsive React frontend demonstrating end-to-end Full-Stack AI engineering excellence.

2. Architectural Decisions & Technical Justifications

Component	Technology Selected	Technical Justification
Agent Paradigm	**Hybrid Tool Calling (ReAct) + RAG**	**Pure RAG fails on math & inventory:** LLMs hallucinate stock counts and promotional calculations. **Pure SQL fails on company policies:** Cannot interpret unstructured legal clauses and returns. The hybrid architecture allows the LLM to deterministically query SQL for numbers and semantically query Qdrant for policies.
LLM Provider	**Google Gemini (Gemini 3.1 Flash Lite / 2.5 Flash)**	Fast inference latency (< 1.5s), native schema-based function calling, excellent adherence to system prompts and Portuguese Brazilian nuances, and robust rate-limit resilience.
Vector Store	**Qdrant (Embedded Mode + Docker Server)**	Qdrant provides state-of-the-art HNSW vector indexing. Using Qdrant's embedded client allows any reviewer to run the application with **zero external dependencies**, while maintaining 100% cloud/server parity.
Embeddings	**Google `gemini-`**	Generates dense 3072-dimensional vector

Component	Technology Selected	Technical Justification
	`embedding-001`	embeddings, delivering outstanding semantic retrieval accuracy for Portuguese legal and operational texts.
Database & Storage	**SQLite + SQLAlchemy**	Zero configuration required for local evaluation. Persists relational tables (Categories, Products, Customers, Orders, Promotions) and conversation memory (Sessions and Messages). Easily swappable to PostgreSQL.
Frontend UI	**React 19 + Vite + Tailwind CSS**	Highlights full-stack capabilities with real-time tool inspection badges (showing evaluators exactly when and why tools are triggered), quick suggestion chips, and responsive UX.

3. Deterministic Tools & Ingestion Pipeline

3.1 Relational Data Ingestion (`data/*.csv`)

All six CSV files were parsed, validated, and normalized into relational tables with foreign-key integrity:

- `categories` (9 music instrument categories)
- `products` (65 items with prices, specifications, and active/discontinued statuses)
- `promotions` (25 seasonal campaigns, with `is_active` boolean tracking)
- `orders` and `order_items` (tracking codes, fulfillment dates, delivery estimates, cancellation notes)
- `customers` (identifying contact information and delivery locations)

3.2 Policy Extraction & Semantic Vector Chunking (`políticas_da_loja.pdf`)

The internal manual was segmented into 10 structured chunks aligned with company

chapters (About Us, Scope Limits, Operating Hours, Payment & Installment Terms, Returns & Remorse, Shipping & Delivery, Promotions, WhatsApp Tone, Warranty, LGPD). These chunks were embedded via `gemini-embedding-001` and stored in Qdrant with Cosine distance metric.

3.3 Agent Tool Definitions

The Gemini agent has access to five deterministic tools:

1. `search_store_policies(query: str)`: Semantic search across Qdrant vector database for business rules, returns, and hours.
2. `query_products(category_name, max_price, min_price, search_text, in_stock_only)`: Dynamic SQL filtering with automatic 5% PIX discount calculation and active promo detection.
3. `get_product_details(product_name_or_id)`: In-depth technical specification lookup, current stock, and detection of past expired promotions.
4. `check_order_status(order_id, tracking_code, customer_identifier)`: Real-time query of order status, delivery dates, internal dispatch notes, and item breakdown.
5. `get_store_info()`: Instant retrieval of physical store address, WhatsApp contact, and scope rules.

4. Guardrails & Persona Engineering

Strict Scope Guardrail (The Golden Rule):
Empório da Música exclusively sells musical instruments. **The store does NOT sell accessories** (strings, picks, cables, straps, cases, pedals, or standalone amplifiers).
The agent strictly enforces this boundary: inquiries regarding accessories are warmly acknowledged, the scope limit is explained, and the customer is courteously redirected to local music accessory shops in Campo Grande.

- **Truthfulness Policy:** The agent never hallucinates pricing, stock availability, or order statuses. It is mandated to query the system before stating numbers.
- **Expired Promotion Transparency:** If a customer mentions an old promotion (such as

the expired 12% Takamine discount), the agent transparently clarifies that the seasonal campaign concluded, presenting current table pricing and standard payment conditions.

- **Consumer Defense Code (CDC):** Accurately enforces the 7-day remorse return window for remote purchases, store-funded reverse logistics, and 10-business-day refund timelines.

5. Evaluated Test Scenarios & Transcripts

Six distinct real-world scenarios were executed and recorded against the live system. Full transcripts are documented in `examples/conversations/`:

Scenario 1: Product Catalog with Price Filter

Prompt: "Quais opções de violões disponíveis custando até R$1000?"

Tool Executed: `query_products(category_name='Violões', max_price=1000)`
Result: Agent returned a structured markdown comparison of available in-stock guitars (Tagima Memphis AC-39, Rozini RC-104, Shelby SN-100, Giannini GF-1R, Yamaha C40, Giannini GN-15), detailing table prices, 5% PIX discounted prices, string types (nylon vs. steel), and 12x credit card installment terms.

Scenario 2: Physical Store Address & Hours

Prompt: "Qual o endereço da loja física e quais os horários de funcionamento no sábado e na semana?"

Result: Accurately informed: Rua 14 de Maio, 3200 — Centro, Campo Grande - MS; Monday–Friday 09:00 to 18:00; Saturdays 09:00 to 13:00; Sundays and holidays closed.

Scenario 3: Specific Product Price & Expired Promotion

Prompt: "Quanto custa o Takamine GD20? Vi em algum lugar que tinha promoção dele, ainda está com desconto?"

Tool Executed: `get_product_details('Takamine GD20')`

Result: Identified product #95 (Takamine GD20 Dreadnought Natural, R$ 2.199,00; R$ 2.089,05 on PIX; 5 units in stock). Crucially, the agent identified that promotion #4 ("Oferta Especial", 12% off) had `is_active=0` (expired), explaining with total transparency that the seasonal promotion had ended.

Scenario 4: Return Policy (Remorse / CDC)

Prompt: "Comprei um instrumento pelo WhatsApp faz 4 dias e me arrependi da compra. Posso devolver? Como funciona o frete e o reembolso?"

Tool Executed: `search_store_policies('devolução por arrependimento prazo frete reembolso')`

Result: Applied Section 4.1 of the policy manual: confirmed eligibility (within 7 days of delivery), required original packaging and accessories, confirmed that reverse shipping is 100% covered by the store, and stated the 10-business-day refund timeframe.

Scenario 5: Real-time Order Tracking

Prompt: "Olá! Gostaria de acompanhar a entrega do meu pedido número 1. Qual o status e o código de rastreamento dele?"

Tool Executed: `check_order_status(order_id=1)`

Result: Found Order #1 for customer Maria Silva (Martin D-28 Dreadnought Natural, total R$ 11.499,00 paid via PIX, status "Delivered", tracking code `BRAB1234567BR`).

Scenario 6: Out-of-Scope Accessory Refusal

Prompt: "Vocês vendem encordoamento de aço para violão e palhetas avulsas?"

Result: Guardrail triggered. The agent warmly explained the store's strict specialization in instruments, declined the accessory sale, and recommended partner stores in Campo Grande.

6. AI Coding Assistant Workflow

In full transparency according to Artefact's evaluation criteria, I utilized **Google Antigravity / Gemini** as a pair-programming partner. The collaborative division of labor:

- **Architectural Strategy:** Formulating the hybrid deterministic SQL + semantic vector RAG architecture to eliminate numerical hallucination.
- **Typing & Compatibility Debugging:** Diagnosed a subtle Python 3.14 typing edge case where the Google GenAI SDK type checker rejected integer inputs into `Optional[float]` parameters, successfully resolving it via `Union[int, float, None]`.
- **Automated Testing & Scaffolding:** Accelerated Pytest fixtures, relational database seeding, and React 19 UI component scaffolding.

7. Known Limitations & Production Roadmap

- **Hybrid Lexical Search:** Combine dense Qdrant embeddings with sparse BM25 (Reciprocal Rank Fusion) for alphanumeric product codes.
- **Multi-Agent Hierarchy:** Decompose duties into dedicated subagents (Sales, Logistics, Policies) orchestrated via LangGraph.
- **Distributed Infrastructure:** Transition SQLite to Managed PostgreSQL with PgBouncer, and local Qdrant to Qdrant Cloud.